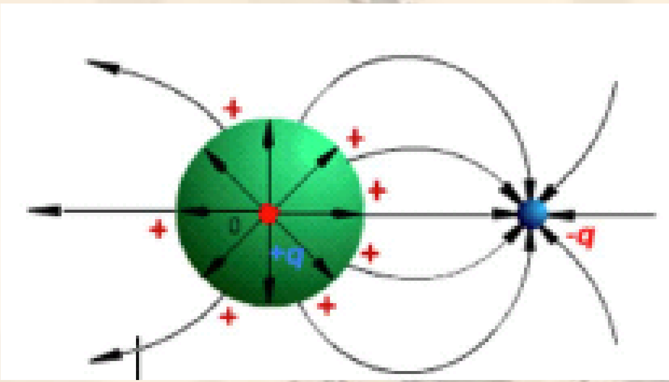
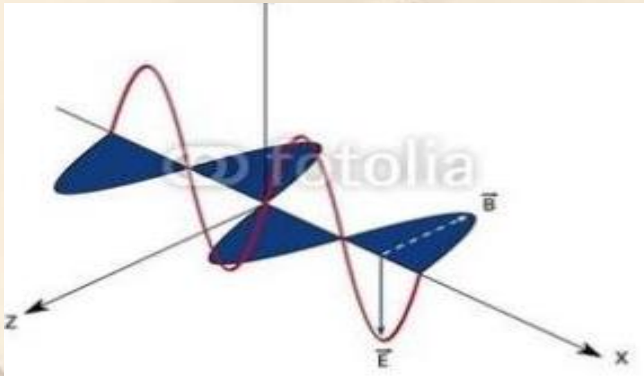
Electromagnetic Field and Microwave Technology





Wuqiong Luo

UoG-UESTC 2026.2

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Join QQ group

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◆ Assignments

- Homework、Online course and Project (10% for project and 5% for homework and attendency) 15%
- Laboratory work based on laboratory reports 10%
- Closed-book mid-term exam (2 hours) 25%
- Closed-book final exam (2 hours) 50%

Spoc online

课堂码：8YHBEW

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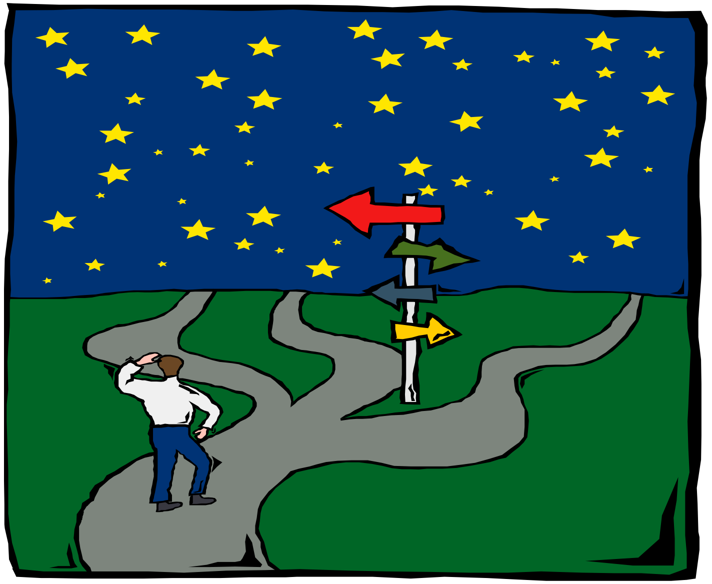
https://www.icourse163.org/spoc/course/UESTC-1468790217?tid=1476748487

About online study

- **Upload your article**
- **A test for one chapter**
- **A final test for online study**
- **Quiz(for rolling)**

Note: finish all above before ddl

Perception
Persistence
Power



- To create with the scientific perception
- To study with the long-lasting persistence
- To overcome difficulties with the power of knowledge

Outline

- Why is this course important?
- Development of EM Theory
- Application for EM Theory
- Contents to be covered
- Course requirements
- Text and reference books

Forces in Nature

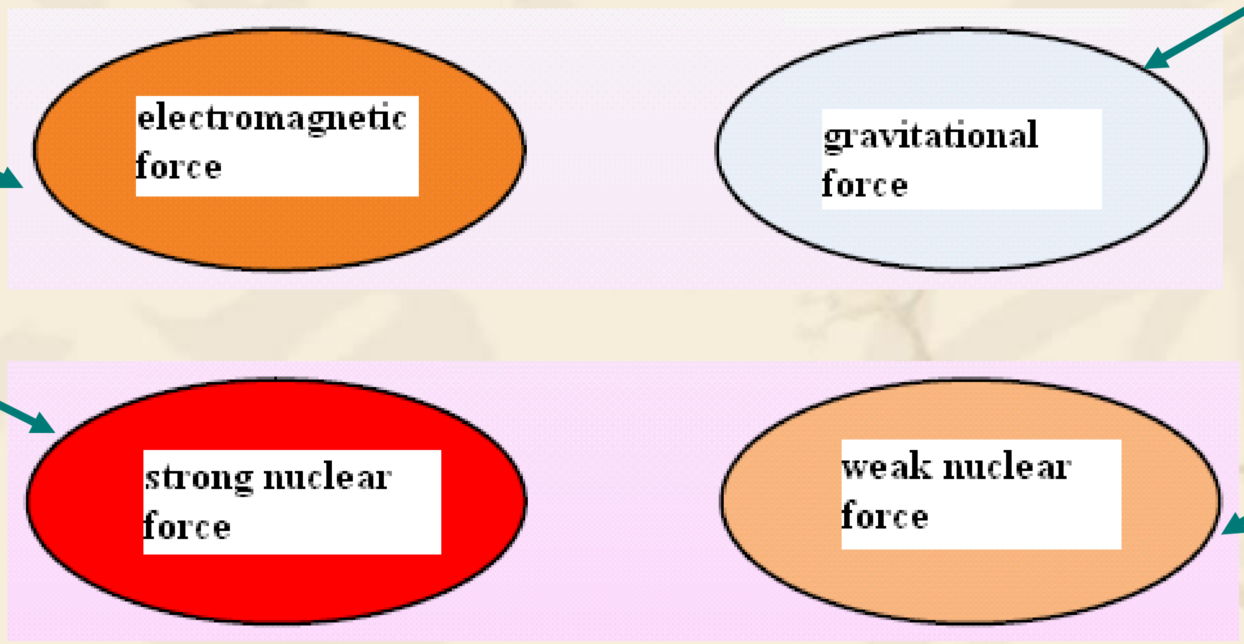
- From measurement, observation and experimentation we know that there are four forces in the Universe:

  electromagnetic force, **gravitational force**,

  strong nuclear force and weak nuclear force.

Electric & magnetic

About an apple

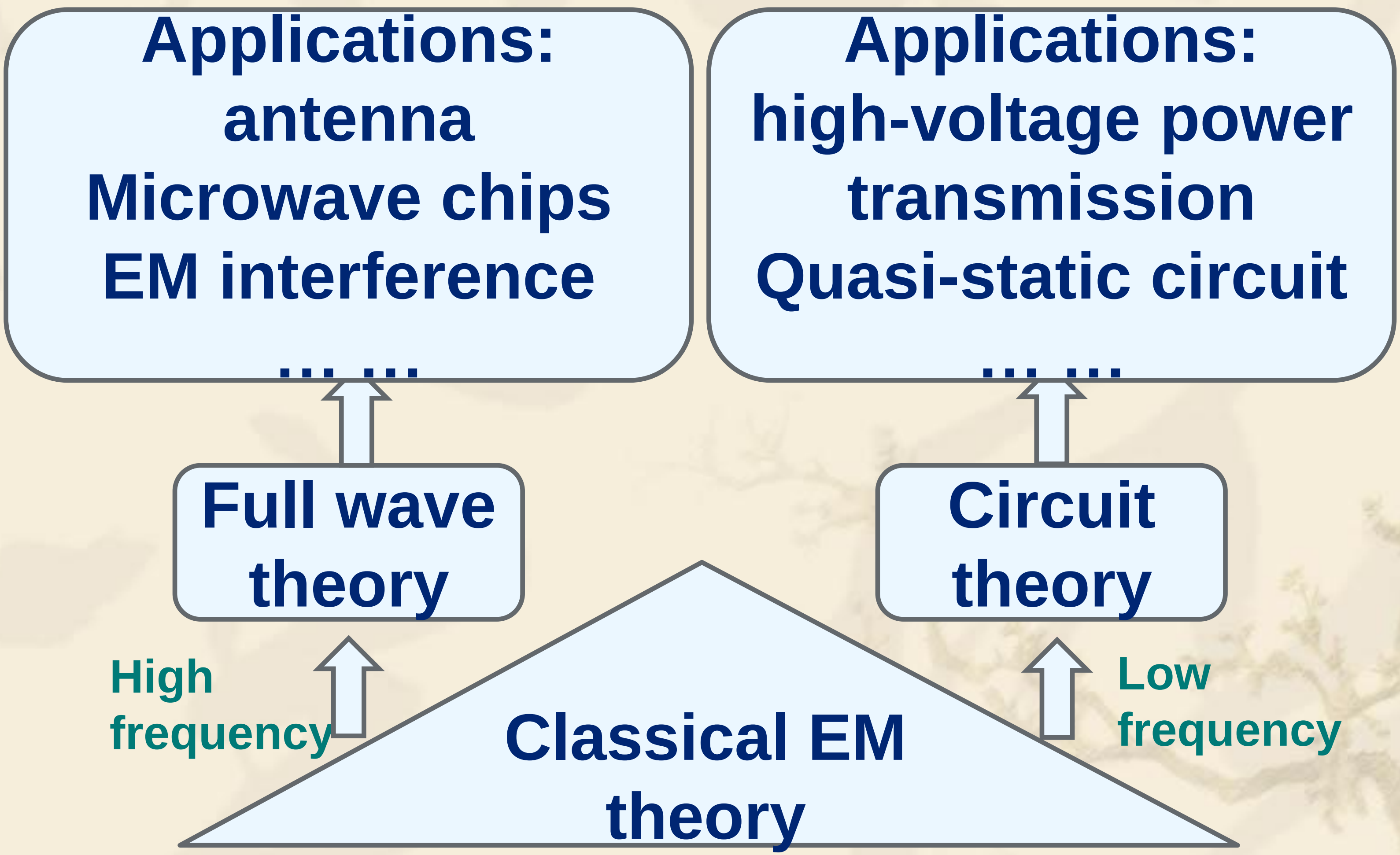


Strongest of the four

10^{-10}~10^{-15}m

Radioactive atom & proton decay

◆ Why is this course important?



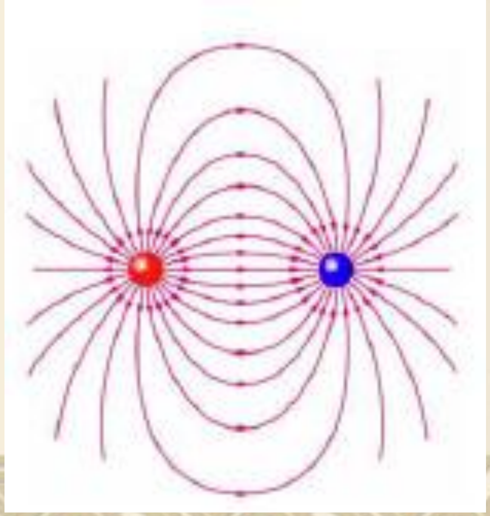
Electromagnetic VS Circuit Theory

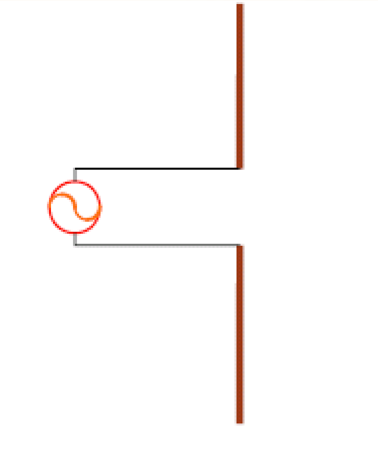
- **Circuit concepts represent a restricted version of electromagnetic concept. A complete electrical network with a closed loop giving a return path for current. Usually, it refers to low-frequency circuit. It's in a *quasi-static state*, which simplifies an electromagnetic problem to a circuit problem.**

E, H	V, I
Most electromagnetic variables are functions of time as well as of space coordinates.	Independent of space coordinates
Vector quantities	Scalar
Governing equations are Partial differential equation	Ordinary differential equation

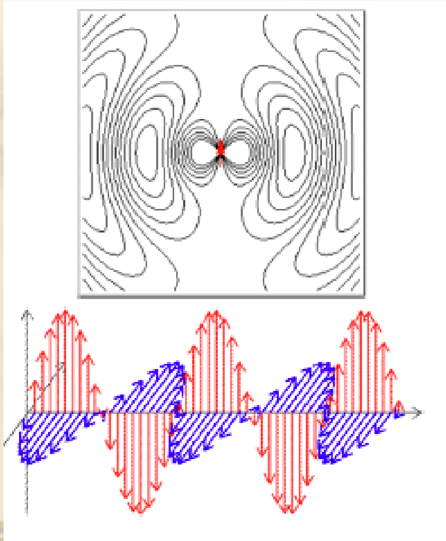
What is Field?

- The behavior of a quantity in a given region in terms of a set of values, one for each point in the region, we refer to this behavior of the quantity as a field (**Force field:** a vector field indicating the forces exerted by one object on another)
- A field is a spatial distribution of a quantity.
- Action-at-a-distance.





A dipole antenna



Wave

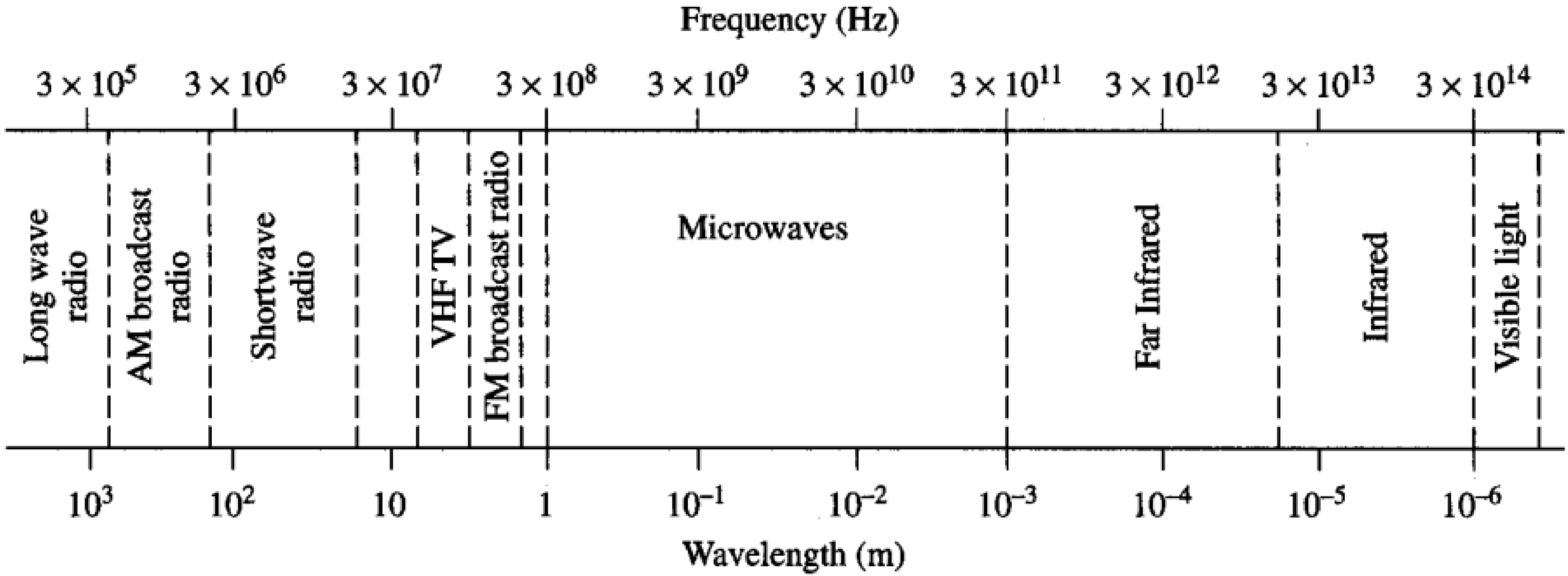
Why is this course important?

- EM field is a quantity existed almost anywhere in human life: Bio-electricity, rays in universe, the currents lighting the light, currents in our computer, etc.
- EM wave is the most widely used carrier for information transmission in the modern society.

- **Basic course for students majoring in electrical and electronic engineering**
- **Essential knowledge for electrical and electronic Engineer**

Why is this course important?

The Electromagnetic Spectrum



Typical Frequencies

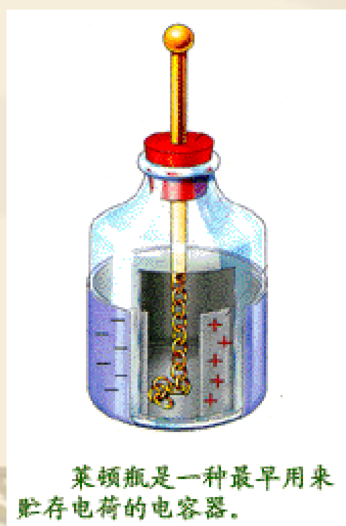
AM broadcast band	535–1605 kHz
Shortwave radio	3–30 MHz
FM broadcast band	88–108 MHz
VHF TV (2–4)	54–72 MHz
VHF TV (5–6)	76–88 MHz
UHF TV (7–13)	174–216 MHz
UHF TV (14–83)	470–890 MHz
Microwave ovens	2.45 GHz

Approximate Band Designations

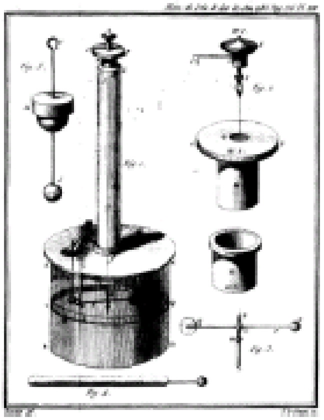
L-band	1–2 GHz
S-band	2–4 GHz
C-band	4–8 GHz
X-band	8–12 GHz
Ku-band	12–18 GHz
K-band	18–26 GHz
Ka-band	26–40 GHz
U-band	40–60 GHz

◆ Development of EM Theory

- **Electrical and magnetic phenomenon is the most important phenomena in nature. The earliest scientists concerned and studied these physical phenomena include *Leiden, Franklin, Volt,* and other scientists.**
- **Before nineteenth Century, the phenomenon of electricity and magnetism has been widely studied as two independent physical phenomena (statically state).**

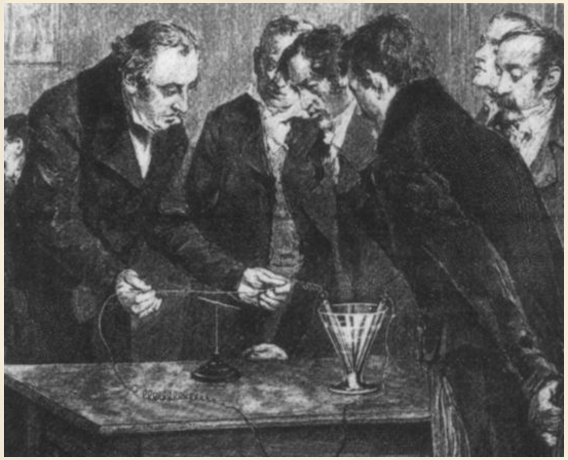


Development of EM Theory

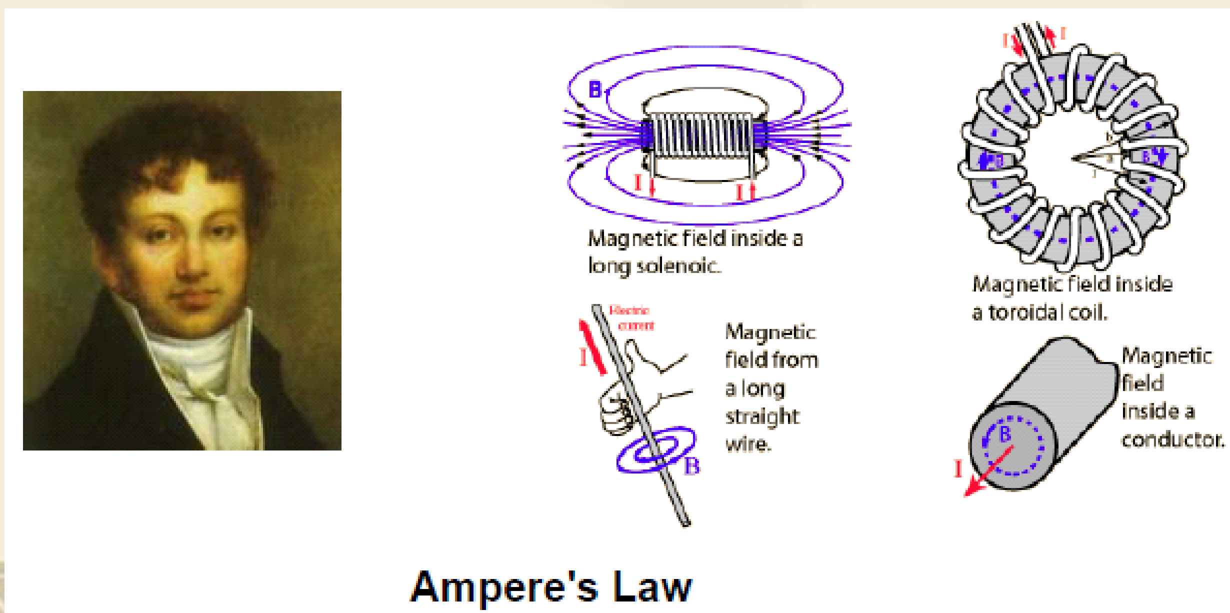


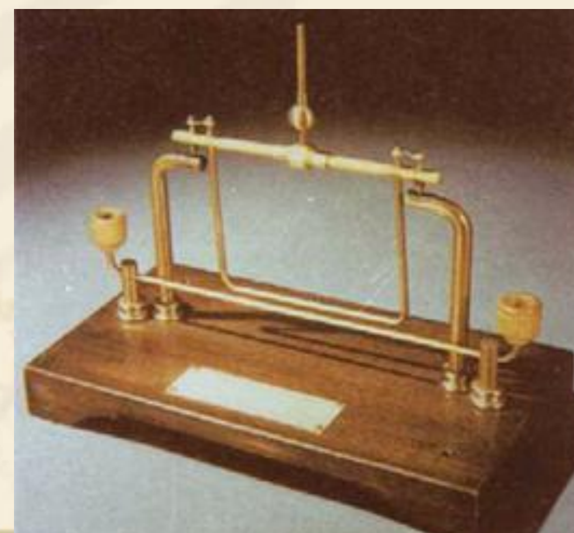
Coulomb's torsion balance

- Magnetic effect of electric current-Oersted 1820

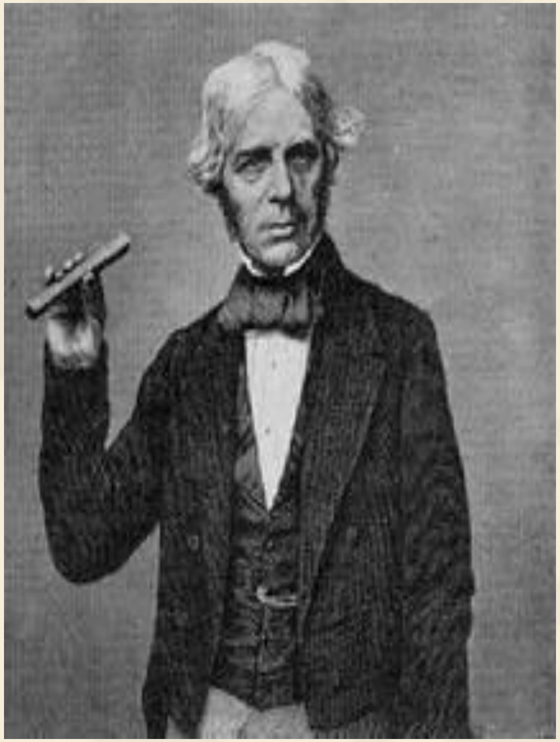


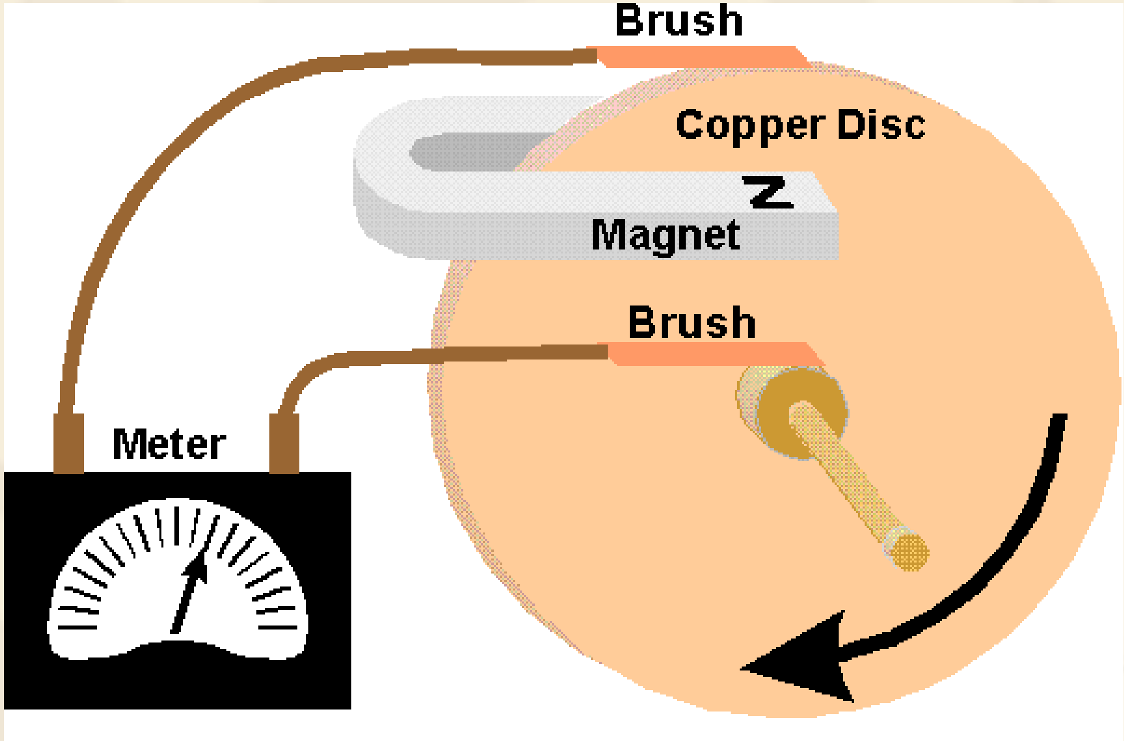
- Ampere's circuital law- Ampere 1820





- Electromagnetic induction law-Faraday 1831

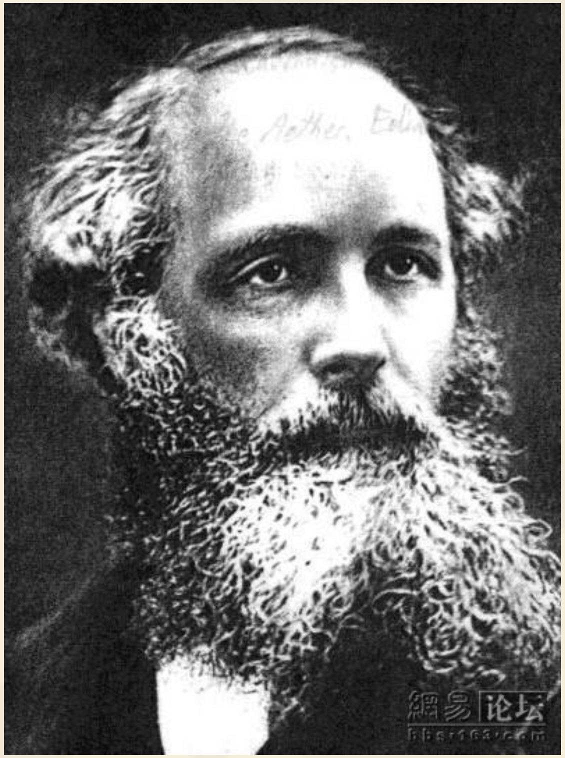


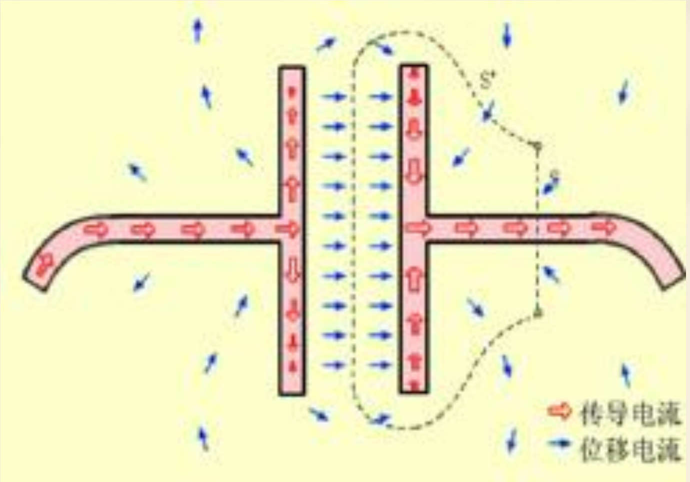


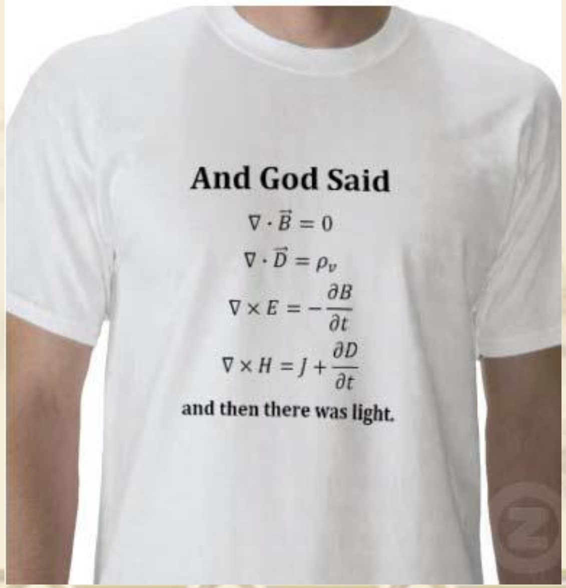
Symbolic illustration of the Faraday Dynamo

One of the greatest experimental physicists in history.

❖ Maxwell's equations-Maxwell 1873

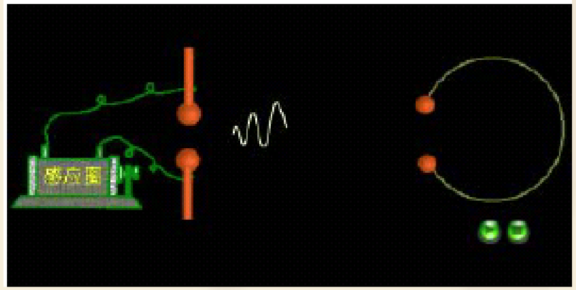


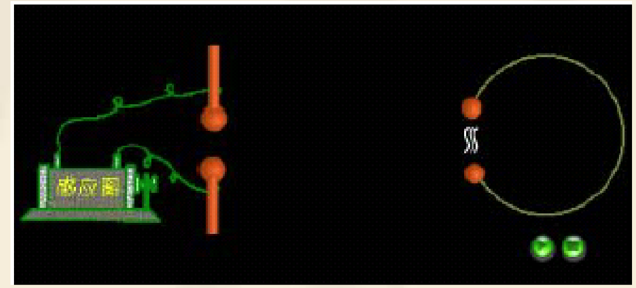




One of the greatest theoretical physicists in history.

❖ Hertz's Classic Experiment- 1887





8 years after Maxwell's death, his theory was perfectly verified.

Application for EM Theory

- **Communication systems**
  -telegraph, telephone, Radio broadcasting

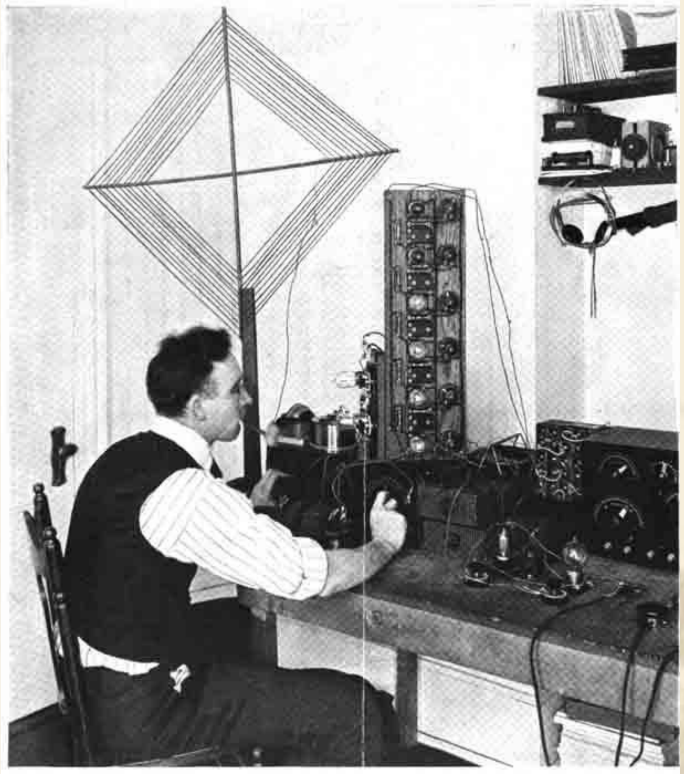
- **G. Marconi- telegraph**
  - **In 1895, telegraph signal was sent over 2.5km.**
  - **In 1899, telegraph signal was sent over the english channel.**
  - **In 1901, telegraph signal was sent over the Atlantic in 3200km.**

Application for EM Theory

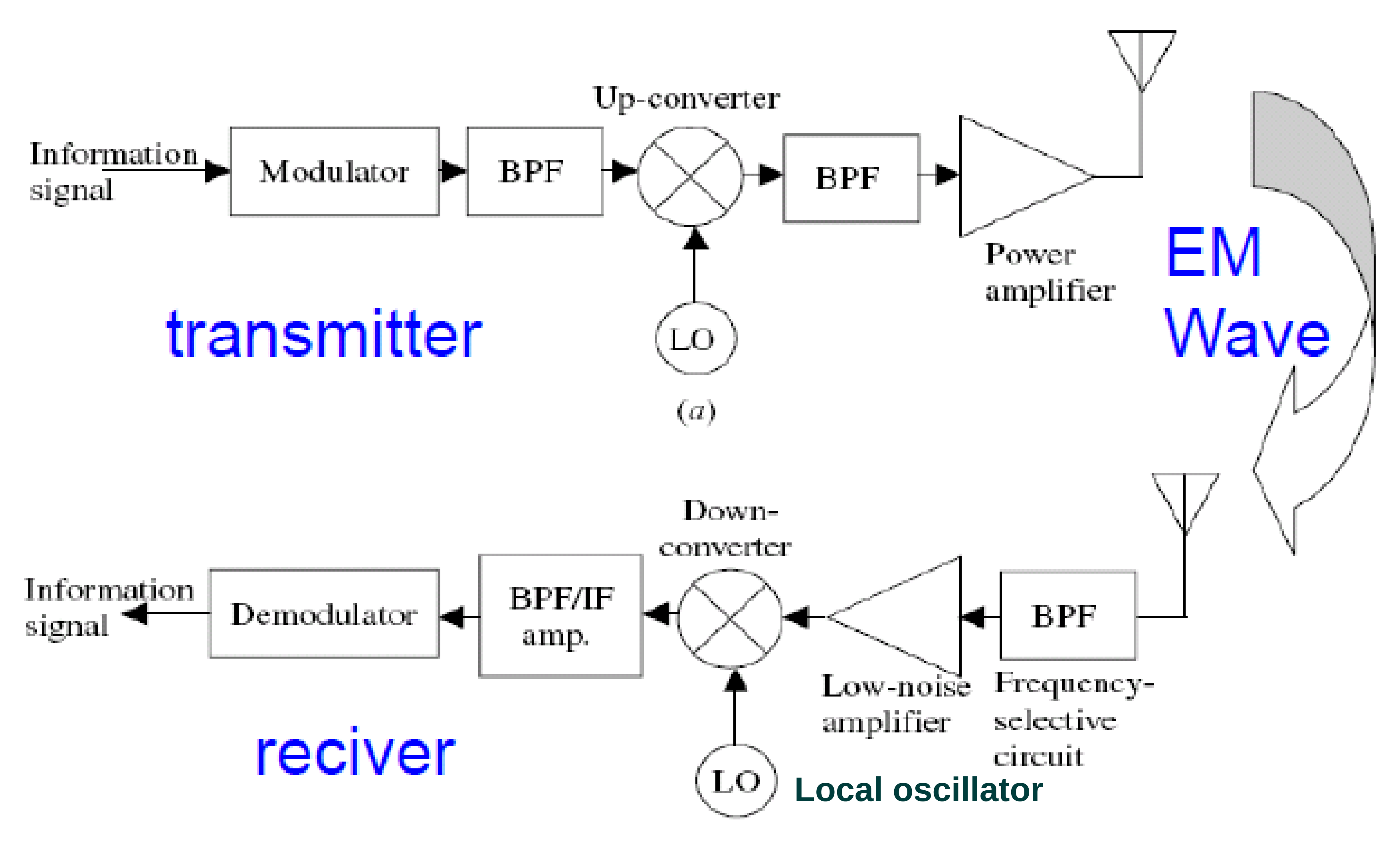
- **Communication systems**

  -telegraph, telephone, Radio broadcasting

- **Fessenden- broadcasting**
  - **In 1906, a radio operator heared the music broadcasting (modulated by a microphone) sent from Boston.**
  - **In 1919, the first radio station regularly braodingcasting language and music program was established in England .**



Wireless Communication System



❖ Communication Technology

Application for EM Theory

- **Communication systems**

  -TV

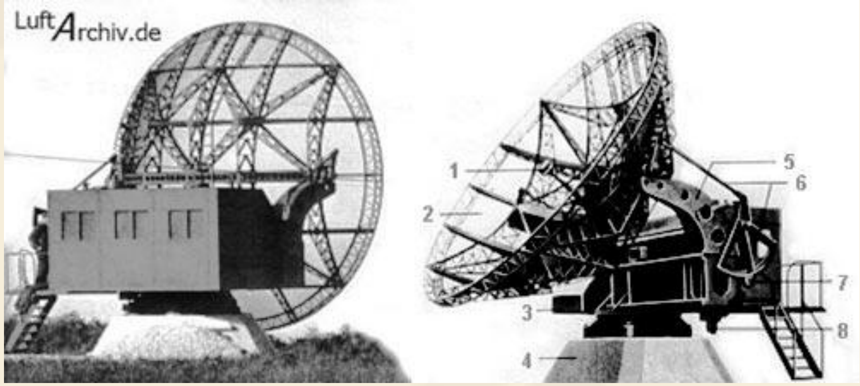
- **In 1884, Nipkow (German) proposed the concept of mechanical scanning television.**
- **In 1927, Baird (Scotland) sent images from London to Glasgow by telephone line.**
- **In 1923 and 1924, Zworykin invented camera tube and picture tueb in succession.**

Baird

Application for EM Theory

❖ **Radar systems**

- **In 1935, Watt invented the warning radar.**
- **In 1938, the first fire control radar was invented in U.S.A.**
- **In 1944, tracking radar was invented.**
- **In 1945, moving target indicator radar was invented.**

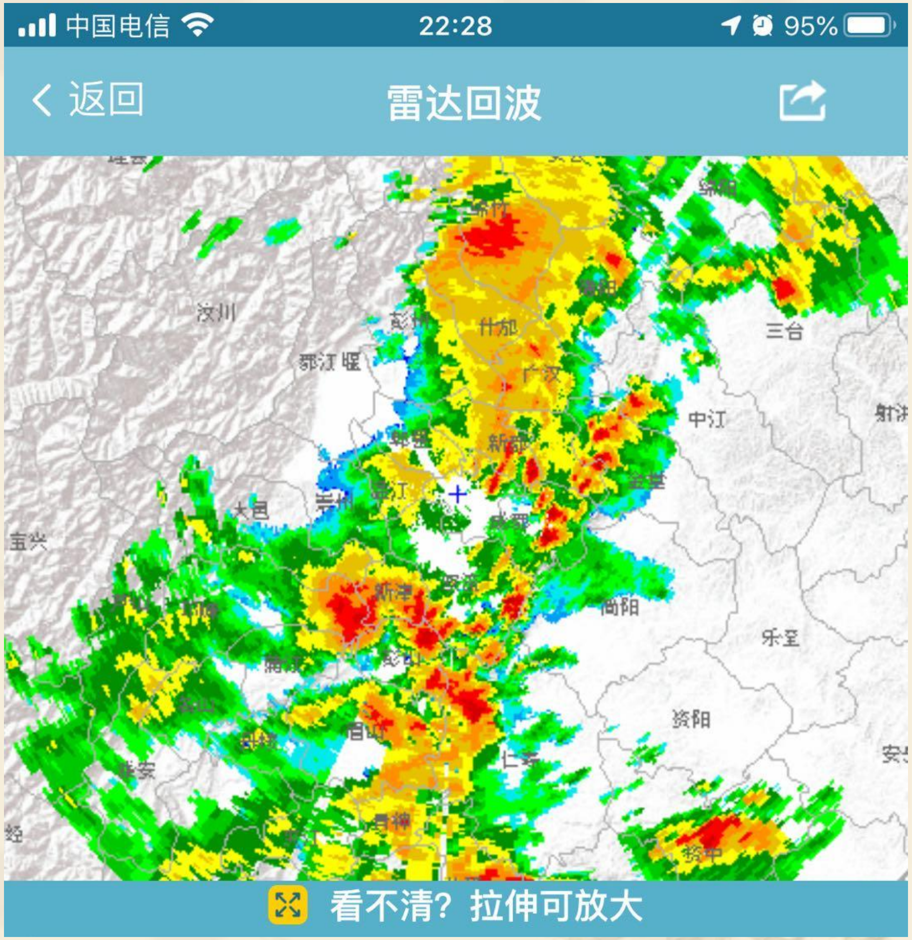


Application

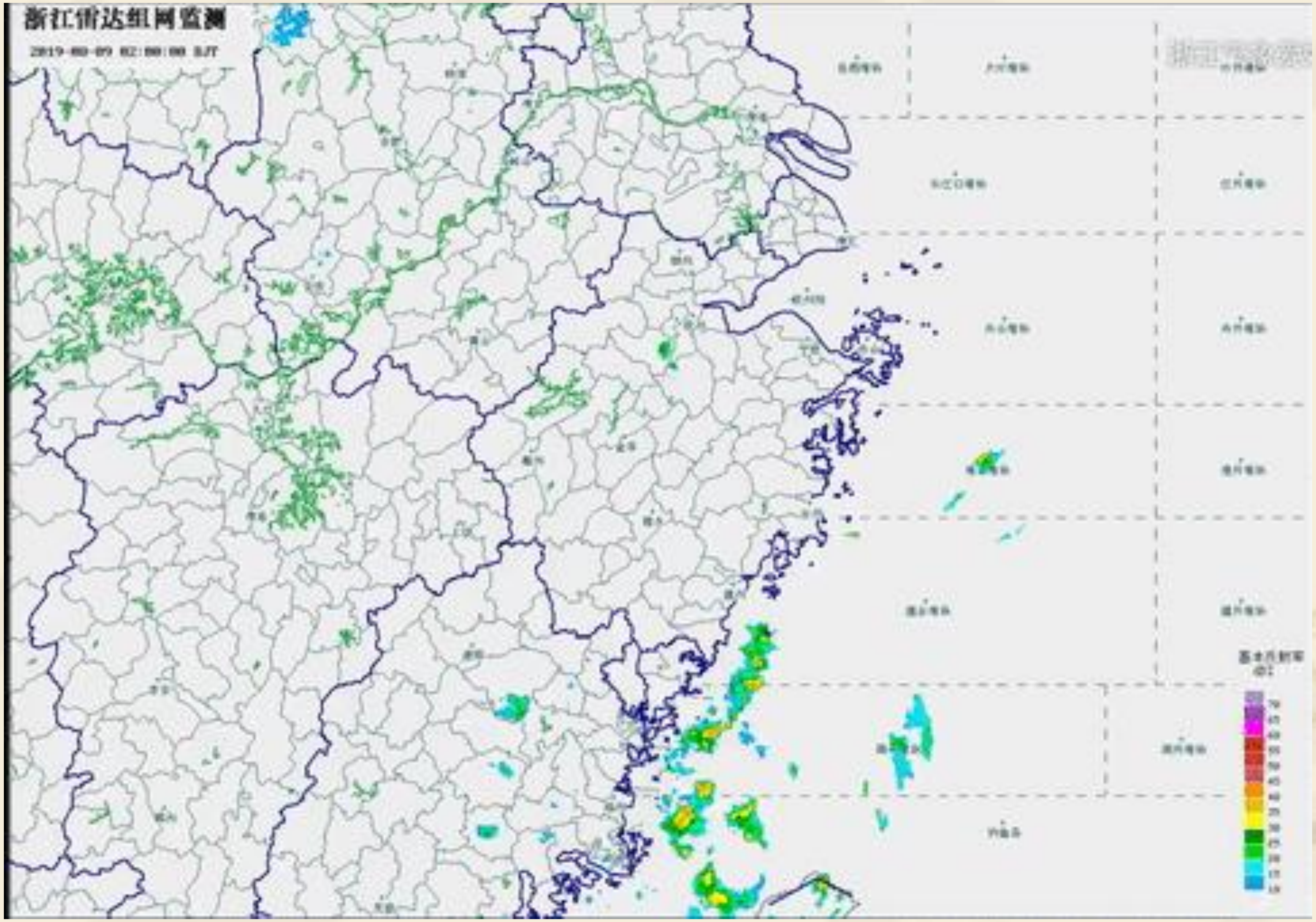
- **Meteorological radar**

A remote sensing device that transmits and receives microwave radiation for the purpose of detecting and measuring weather phenomena; includes Doppler radar, which is used to determine air motions (to detect tornadoes), and multiparameter radar, which provides information on the phase (ice or liquid), shapes, and sizes of hydrometeors.

Base Reflectivity: blue-purple (10-70dBz**)**
Rainning fall intensity:
E.g. yellow-----10mm/h
red ---------20mm/h



Super Tyhoon Lekima (2019)



Applications

- **Biomedical Researches**

An important topic of discussion today is the impact of radio frequency electromagnetic fields on human tissue. People are interested in:

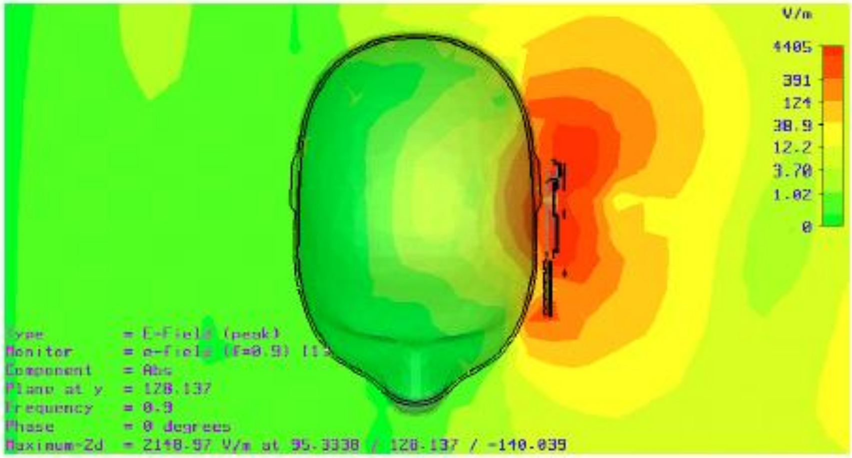
- the Specific Absorption Rate (SAR) in the entire head,

- the 1 and 10 gram averages of the SAR values including the peak value locations,

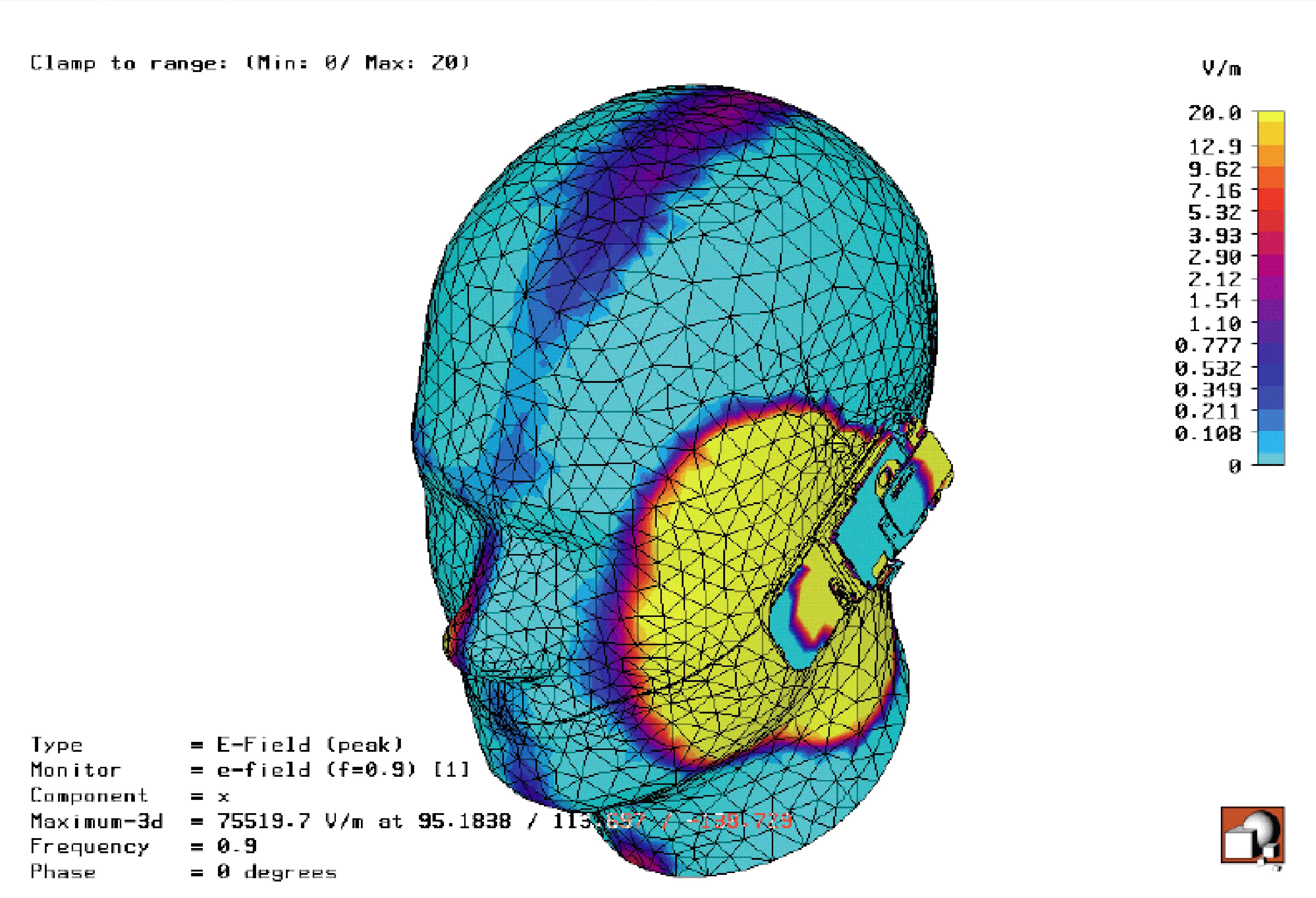
- the whole-body average SAR throughout the head,

- conduction currents on the outer surfaces of the head,

- and various antenna characteristics including input impedance and efficiency.



Application for EM Theory



Clinical applications

- **SmartPill（美敦力）**

  测量：**胃 / 肠压力、pH、温度、 transit 时间**

  天线：434MHz，传输距离约 1 米

  用途：胃排空、小肠传输、便秘 / 腹泻、胃食管反流评估

- **Bravo pH 胶囊（美敦力）**

  测量：**食管 pH（24–48 小时）**

  天线：434MHz，固定于食管壁

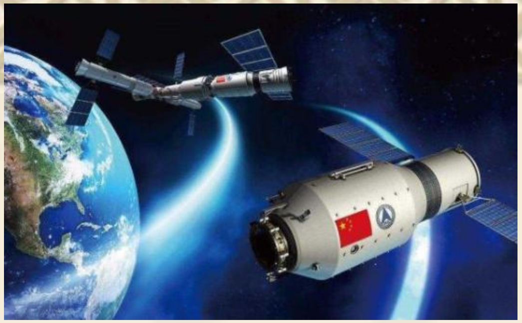
  用途：反流性食管炎、Barrett 食管诊断

Applications

❖ Satellite Communication

- In 1958, the first communication experiment satellite was launched by American.
- In 1964, communication and TV broadcasting bewteen North America, Europe, and Africa were realized with synchronous satellites.
- In 1965, the first synchronous communication satellite is in commercial application.
- In 1970, our first satellite 东方红一号 was launched.

Applications



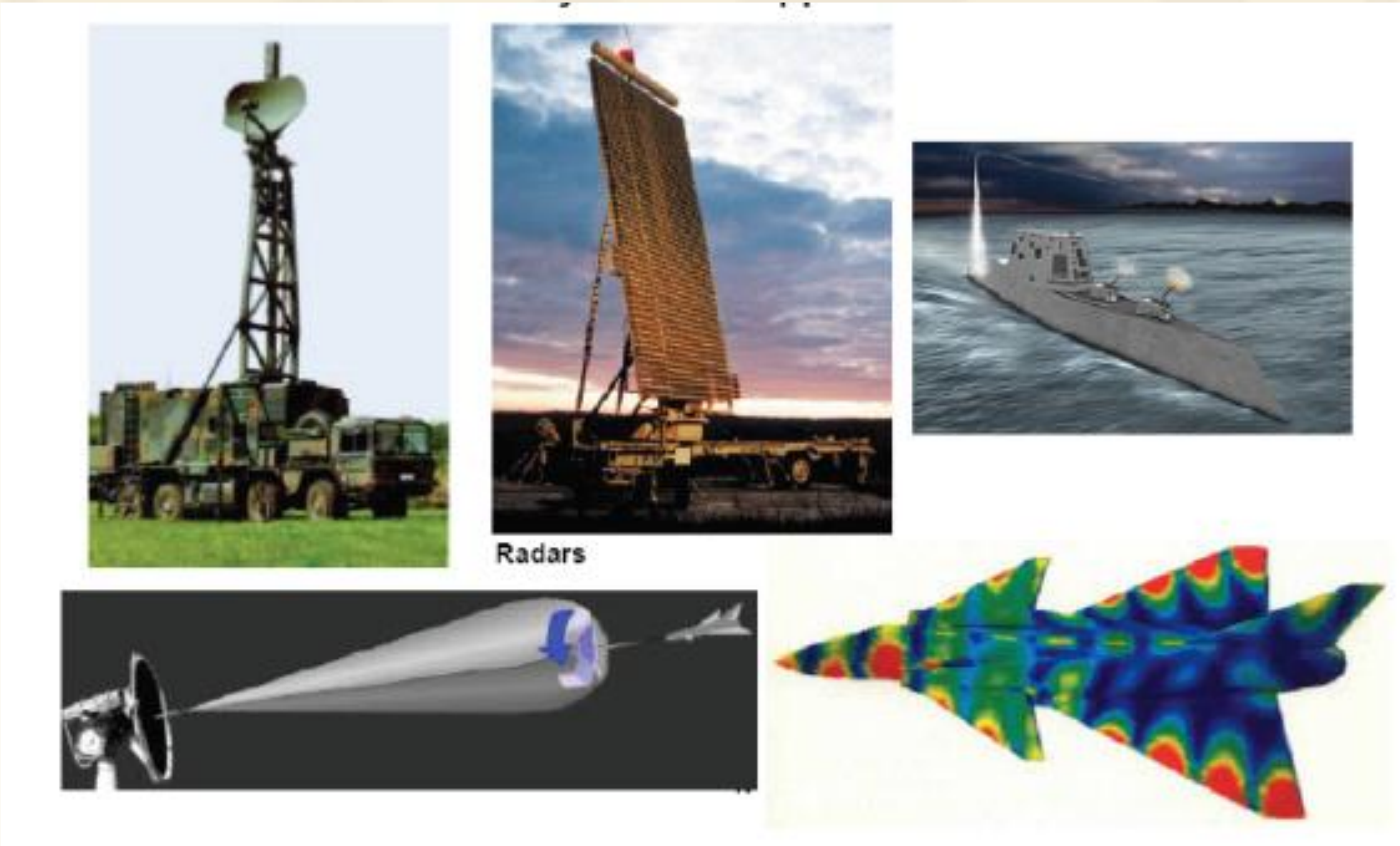
❖ Satellite navigation

- In 1964, the transit navigation satellite system was proposed in U.S.A.
- In 1973, plan of GPS was proposed.
- In 1990, GPS was finished
- Beidou System

项目	中国	美国
发射次数	92 次	198 次
入轨卫星数	约 370–380 颗	约 2378–3718 颗
主力星座	国网、遥感、北斗	星链（Starlink）

Application for EM Theory

- **Military application**



◆ Contents to be covered(68)

Lectures: 56 Experiments: 12

- **Chapter 1 Introducation (1)**
- **Chapter 2 Vector Analysis (6)**
- **Chapter 3 Static Electric Fields (5)**
- **Chapter 4 Solution of Electrostatic Problems (8)**
- **Chapter 5 Steady Electric Currents (4)**
- **Chapter 6 Static Magnetic Fields (6)**
- **Chapter 7 Time-Varying Fields and Maxwell's Equations (8)**
- **Chapter 8 Plane Electromagnetic Waves (12)**
- **Chapter 9 Guided Electromagnetic Waves (2)**
- **Chapter 10 Electromagnetic Radiation and Principle (2)**
- **Exercise class (2)**
- **Experiments (Theory 4 + Labwork 8)**

◆ Course Requirements

Static fields

- **Describe the basic laws of electromagnetic fields, including practical boundary conditions;**
- **Solve electrostatic and magnetostatic problems using the potential function and standard methods such as images and separation of variables;**
- **Relate capacitance to stored energy in electric and inductance in magnetic fields;**
- **Demonstrate general properties of Maxwell’s equations for time-varying fields and derive the Poynting vector;**

◆ Course Requirements

- **Derive the plane-wave solution from Maxwell's equations in free space and non-conducting media;**
- **Analyze reflection and refraction of plane waves at a plane interface;**
- **Show that waves can propagate in transmission lines and waveguides and deduce the properties of guided waves;**
- **Calculate the radiation field of an elementary dipole and state the duality principle for electric and magnetic dipoles.**

Survival Tips

- Attend all lectures (!!).
- Read the corresponding book sections before the lecture
- Solve all the homework problems with **little discussion**.
- Work together to finish the presentations with the specified topics (3 members in one group)
- Do not look up solutions until you have spent enough time thinking!

Lecture: 56 classes

Pay attention to the schedule

Lab: 12 classes (2 times)

Time : to be decided

1. **Understand what is the electromagnetic wave;**
2. **Design and fabricate the electromagnetic wave sensor;**
3. **The propagation characteristics of electromagnetic wave;**
4. **Polarization experiment of electromagnetic wave.**

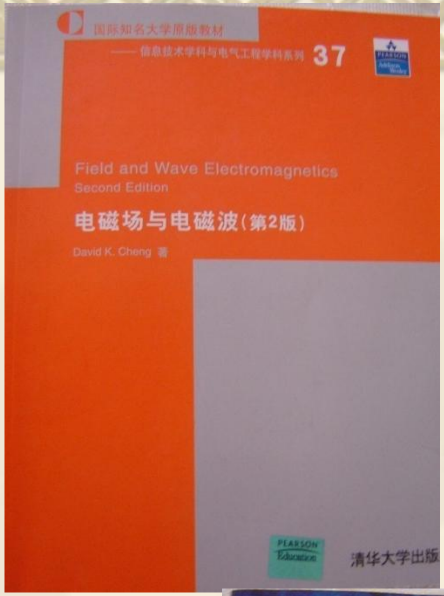
◆ Text and reference books

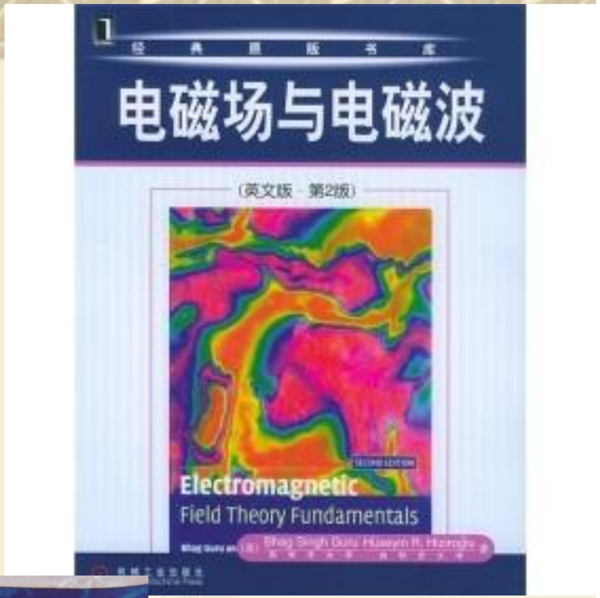
<u>Textbook:</u>

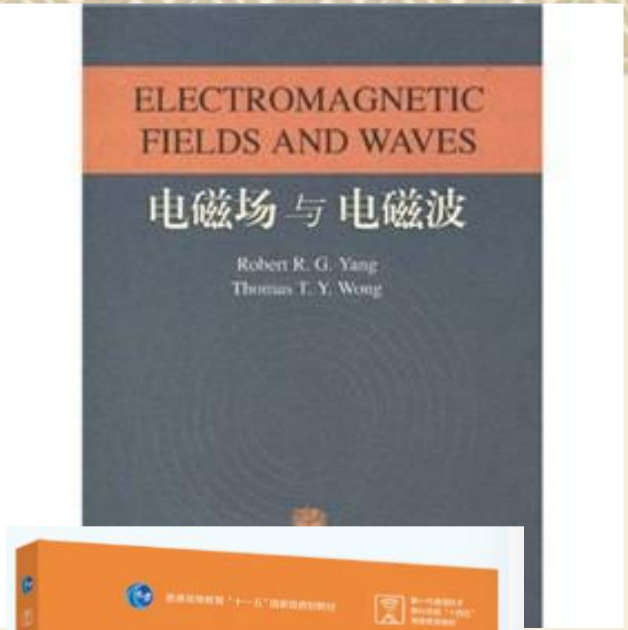
***Field and Wave Electromagnetics*, David K. Cheng, Tsinghua University Press, Second edition, 2007**

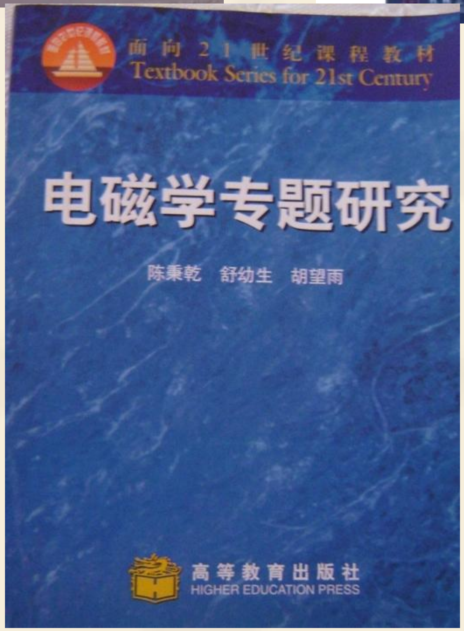
<u>Reference:</u>

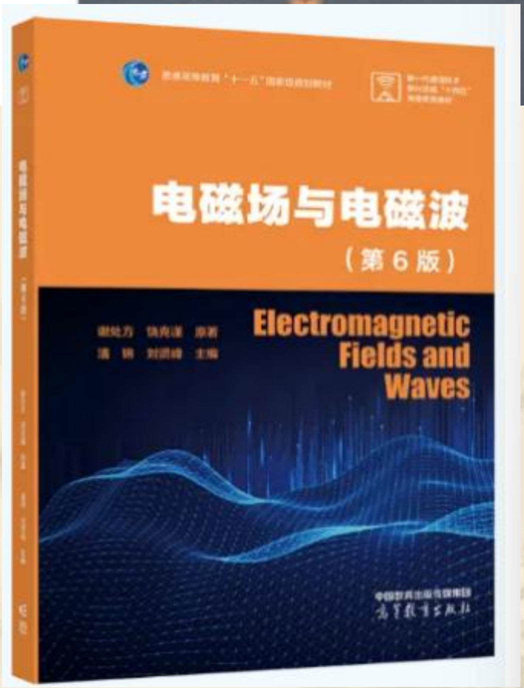
1. **Electromagnetic Fields and Waves, Robert. Yang, Thomas. Wong, Higer Education Press, 2006**
2. **Electromagnetics With Applications. 5nd ed, John D. Kraus, Daniel A. Fleisch. Tsinghua University Press，2001**
3. **Applied Electromagnetism. 2nd ed . Liang Chi Shen, Jin Au Kong . PWS Publishing Company, 1987**











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Mathematical difficulties

- **vector analysis**

  (The direction of vector integration changes with the independent variables)

- **Calculus operation (Linear integral, surface integral, volume integral, limit)**

About home work

- Write a small paper to study on one of the physics or some important discovery/experiment in the development of electromagnetic theories, and give your evaluation.(upload to Spoc， DDL：30th, April)

Hope you enjoy your study ！ ！